Energy & Buildings 203 (2019) 109446

Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enbuild

A study on occupant behaviour related to air-conditioning usage in residential buildings

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ARTICLE INFO

Article history:
Received 4 June 2019
Revised 18 September 2019
Accepted 20 September 2019
Available online 21 September 2019

Keywords:
Air-conditioning
Residential building
User behaviour
Clustering

ABSTRACT

Residential buildings are typically associated with high energy consumption, especially because of air-conditioning (AC) usage. Occupant behaviour can noticeably affect how an AC operates and thereby contribute to the increasing energy consumption. This problem is more pronounced with the split-type air-conditioner that is popular in South China. This study identified three representative patterns related to occupancy and AC on/off settings according to AC operation recordings from 102 bedrooms in several residential buildings of metropolitan Guangzhou, South China. Bedroom AC operation details were recorded using individual smart socket units. The resulting data showed clustering for daily on/off times, the duration of each operation, and AC 'run-over-night' probability. These clusters revealed three user-configuration patterns (i.e. night-time AC usage only (49%), both night-time and noon AC usage (6%), and low AC usage (45%)). Occupant age, total floors, and room area indices were most closely related to occupant behavioural patterns. Results indicated that AC units should be 'turned on' in timed steps when operation rates are higher than 30% or 20% to achieve acceptable thermal comfort performance. These findings should be useful for those intending to reduce overall energy consumption because the analysed schedules were based on realistic usage rather than conventional assumption (i.e. the full-time/full-space usage).

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1. Introduction

Residential buildings consume large amounts of energy throughout the world. In 2014, Chinese residential buildings consumed around 209 megatons of coal equivalent (Mtce) (excluding urban district heating, which accounted for 23% of total building energy usage) [1,2]. This was comparable to the 275 Mtce used by Chinese public and commercial buildings. In fact, energy consumption has nearly tripled among residential buildings since 2001. This is due to urbanisation, increased building area per person, and the popularity of air-conditioning (AC). Residential buildings in the United States consumed 21,640 trillion BTUs of energy in 2018, thus accounting for 21.37% of the country's total energy usage; this was higher than commercial sector usage over the same period (18.38% of the total) [3]. AC and space-heating contributed to 17% and 15% of the total energy consumption among US residential buildings in 2015, respectively; this was highest among all sectors [4]. In addition, residential buildings account for 21% of the total energy usage in Hong Kong as a representative subtropical city, with 25% of this being due to AC usage [5]. It is thus crucial that residential buildings be equipped with energy-saving features. This is especially critical for systems designed to produce thermal comfort, including AC.

Individual residential occupants may remain indoors for different periods of time and thus have different thermal comfort needs [6]. This means that occupant behaviours (especially those related to room and AC usage) significantly contribute to overall energy consumption [7–9]. This is dissimilar to operating practices in most commercial and public buildings, where institutional energy-consumption rules are typically set for occupants over the course of a scheduled day. Here, it is presumable that office workers tend to stay indoors for less time than retired elderly persons and children. However, the latter two may need less cooling while

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https://doi.org/10.1016/j.enbuild.2019.109446
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indoors because of their relatively low metabolism rates. Split-design AC on/off conditions are also fully user-controllable to fit different working statuses and lifestyles [10]. Here, different AC operation periods among individual occupants may lead to high uncertainty about energy usage when attempting to evaluate energy-saving protocols [11–13]. This is a problem because energy saving strategies are needed to confront energy usage issues among buildings with different occupancy patterns. For instance, proper shading and envelope insulation can save energy for buildings with intensive daytime usage [14]. However, these measures may not be effective for structures with high night-time usage. It is thus essential to study different occupant schedules to determine specific AC needs for each building. This will provide useful information when conducting energy evaluations.

Theoretically, the occupant behaviour can impact the energy consumption and the energy saving studies like the outdoor climate condition. The impact of the climate to the AC unit operation mode and the energy consumption can mainly be rendered by the algorithms of many well-developed simulation tools. However, the occupancy and AC operation features for residential buildings of many engineering design codes and standards were determined without sufficient field investigations. For example, local energy codes for buildings in China (e.g. for hot-summer/warm-winter and hot-summer/cold-winter areas) typically set constant indoor air temperatures (e.g. 26 °C) throughout the day [15,16]. This leads to 'always-on' daily AC operation. Another problem is that local design guides do not provide varying occupancy rates for residential buildings based on the time of day. Similarly, the prototype residential building models defined by the International Energy Conservation Code assume 'always-on' AC operation schedules [17]. This 'always-on' assumption is also used by the US Department of Energy for apartment buildings [18]. However, this assumption has limitations, especially in developing countries such as China, where occupants typically terminate power to their AC units when leaving the space to reduce costs and conserve energy. It is thus necessary to investigate the typical occupancy and AC operation features for engineering problems systematically.

The occupant behaviours were vastly studied in previous studies for both residential and commercial/public buildings. The occupancy is being shifting from a pre-defined deterministic input to an outcome of many factors such as the environment condition [19]. In this connection, the occupant behaviour and the environment may affect each other, and should be determined step by step [20,21]. This can be important for the design and operation strategies for the centralized AC systems, and understanding the user responses to the thermal environment [22]. Many sensors are required to recording the thermal status and detecting the occupant activities [23]. However, it can be difficult for using sensors in large-scale studies of residential buildings due to the privacy concerns. In terms of applications, the approach requires specialized user-defined codes to co-simulate with the energy modelling programmes [24], which can be somehow difficult for the general engineering uses. For residential buildings, on the other hand, the commonly used split-type AC units usually operates in an on/off mode. In this connection, the impact of the environment conditions on the on/off behaviour were somehow similar to the thermostats of AC units that can mainly be determined by the simulation algorithms. Therefore, developing deterministic occupancy and AC operation schedules that identifies the typical hours with or without AC needs should be a rational start point to modify the 'always-on' assumption for the residential buildings.

Many recent works have proposed deterministic occupancy and AC usage patterns for residential and commercial buildings. Some of them gain information by questionnaire responses [25,26], and the others monitor the AC energy consumption according to the power meters [27,28]. The indirect measurement approaches were used probably due to the privacy concerns. However, the data on occupancy status acquired by these questionnaires are less accurate than direct measurements. The energy usage, on the other hand, implicates the operation status of compressors more so than the occupant behaviour. The energy users, in many studies, were systematically clustered into several groups, and the activities were revealed in detail [29,30]. However, the social and financial features of the group members were rarely studied, probably because the occupants are reluctant to give the relevant information. In addition, the field investigations based on many citizens determines the probability of the AC operation, which should be converted to specific on or off conditions of the AC units for routine engineering simulations. In previous studies, the AC was 'turned on' when its operation rate was over an intuitive threshold (e.g. 0.5) [31]. However, it is not clear whether occupants would be satisfied with their thermal environments under this logic of operation. It is thus necessary to study the typical AC usage features of the occupants, and determine the occupancy and AC on/off schedules accordingly for engineering simulations.

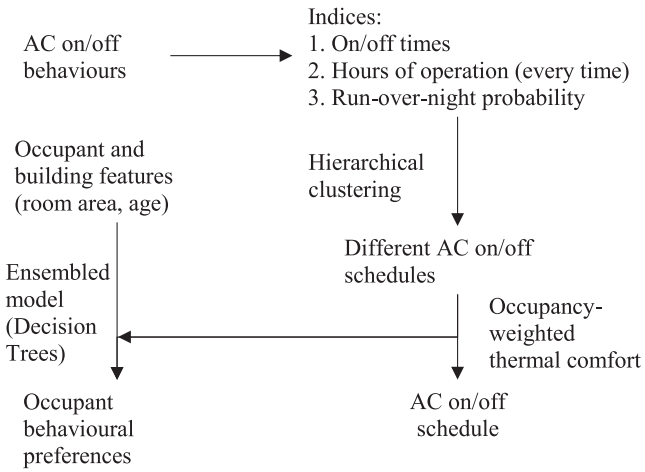


Fig. 1. Research procedure.

This study examined occupant behaviour and AC operation features among 102 bedrooms of a residential building under hot and humid summer conditions in Guangzhou, China. On/off status was detected using individual Zigbee smart sockets. All resulting information was thus directly related to individual user behaviours. Fig. 1 provides a brief overview of the research procedure. AC operation behaviours were summarised as indices (including daily on/off times, duration per operation, and overnight probability indices for each bedroom). Individual room users were thus clustered into representative occupant types with distinctive AC operation rate schedules. Each type was correlated to personal features such as salary, age, and workplace distance from home using bootstrap aggregating decision trees. The importance of each feature to its respective correlation was also quantified. Finally, we tested occupancy weighted thermal comfort by turning on AC units at different operational probabilities to determine AC on/off schedules for use in simulations.

2. Materials and methods

This section specifies the details of the AC on/off measurements, the room and the occupant features (2.1). The days without any AC usage due to the user willingness (2.2) were identified. Three variables that specify the user behaviour, i.e. on/off actions per day, duration of each operation, and run-over-nigh rate, were constructed and studied for the subsequent clustering (2.3).

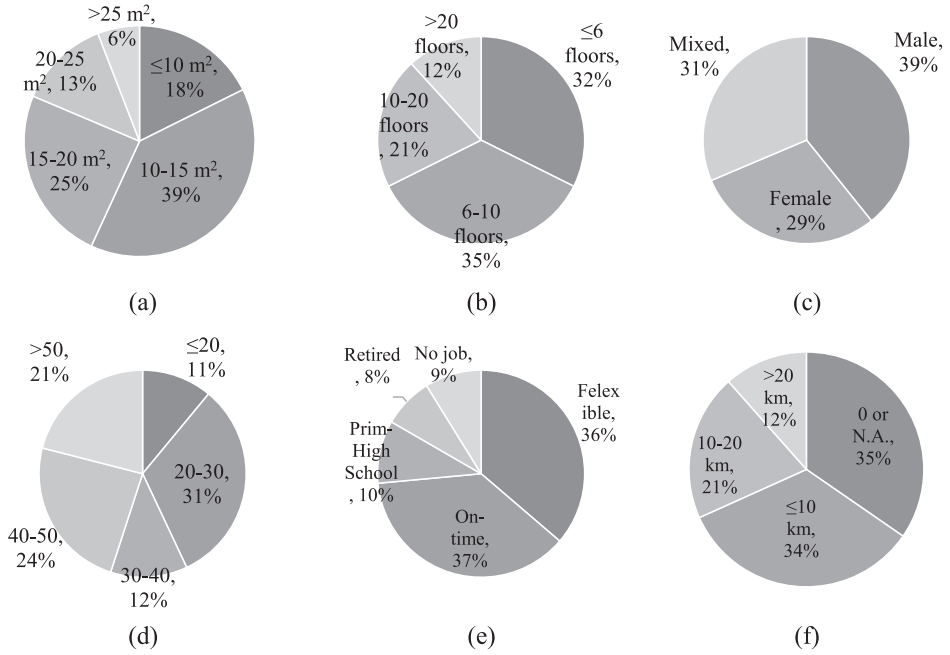


Fig. 2. Percentages of all 102 sampled bedroom features: (a) Total floors, (b) window direction, (c) user gender, (d) user age, (e) user schedule flexibility, and (f) user travel distance for work.

2.1. Data collection

Data on AC operation were collected in Guangzhou, China (113.27E, 23.13N), which has a humid subtropical climate. That is, summers are hot and humid, while winters are generally mild and relatively dry. This climate leads to substantial AC cooling needs that result in heavy AC energy consumption during summer months. Data were thus collected from 1 July to 31 July in 2018, which is generally the hottest period of the year and therefore when AC cooling needs are highest. Data on AC operation were collected from 102 bedrooms of both medium and high-rise buildings in urban Guangzhou. Fig. 2 shows the essential features of all 102 bedrooms, including window directions, building heights, occupant ages (bedroom users instead of holders), and work distances from home; each of these factors may influence bedroom occupancy schedules. Fig. 2(a) shows that most sampled room areas were 10–15 m^2, while around 25% of the samples were 15–20 m^2, and around 18% were less than 10 m^2. Fig. 2(b) shows that one-third of the samples were from low-rise building with six or fewer levels, which were typically built at earlier times than buildings with 10 or more levels. The 'mixed' sector shown in Fig. 2(c) indicates that there were two persons in a given room. Fig. 2(d) shows that the test covered a broad range of occupant ages. About 30% of all occupants under evaluation were aged from 20 to 30, while around 25% were from 40 to 50, 20% were over 50, and around 10% were either less than 20 or between 30 and 40. Fig. 2(d) shows that fixed-working-period and student occupants accounted for 37% and 10% of the total, respectively, while the remainder had flexible schedules and may have thus stayed in their bedrooms for longer periods of time. Fig. 2(e) shows that around one-third of the sample worked within 10 km from their residences, while 35% did not go out for work, and the rest travelled more than 10 km for work.

2.2. Data quality evaluations

We identified AC on/off status according to real-time power consumption as detected by the individual Zigbee smart sockets. AC units were defined as 'in operation' when power consumption is greater than 2 watts. Otherwise, AC units were considered 'out of work'. The smart socket reads the real time power consumption data that were uploaded to a program. Then the operation period was manually read in a minimum time interval of 1 min. It was inevitable that there can be days without AC usage recordings though we tried to keep the measurement uninterrupted. Fig. 3 plots the occurrence frequencies (FOC) of the AC users that were featured with 0–20 days without AC usage recordings. It shows that there were at least one AC on/off operation per day for around 20% of the 102 AC users. For most (80%) of the users under investigation, there were less than 12 days without any AC usage recordings. The no-AC days can be explained as internet connection problem, user being away, or the user willingness. It was important to systematically identify whether the no operation was due to the user willingness or not.

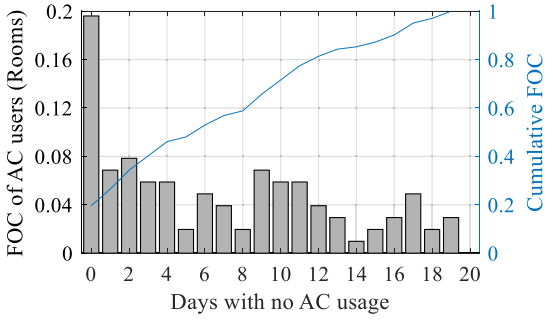


Fig. 3. The occurrence frequencies (FOC) of AC users for 0–20 days without AC usage.

However, monitoring the occupancy of the homes directly using cameras or occupancy sensors can be difficult due to the confidential concerns of the users. In this connection, it is assumed that the habit of a user, or the user's reliance on AC, should not change during the measuring period. It is thus rational to determine whether the no-AC-usage was the user's willingness or not according to the other days when there are any AC operation recordings. In other

Table 1
Definitions of the constructed variables.

Constructed variables	Definitions of the variables
On/off actions per day	Ratio of the total on/off times[1] to the number of days[a,b] in the period
Duration of each operation	Ratio of the total operation hours[1] to the half of the total 'on' and 'off' actions[a]
Run-over-nigh rate	Ratio of the days when the AC unit ran throughout the night[a] to the number of days[a,b]

[a] Either on weekdays or weekends;
[b] The days on denominators for each AC users, the days being counted should have at least one 'on' or 'off' action, excluding the 42 defined by $T_1 = 0.4$ and $T_2 = 0.6$ on Fig. 4.

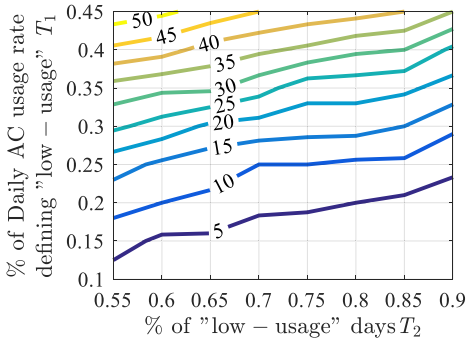


Fig. 4. The occurrence frequencies (FOC) of AC users for 0–20 days without AC usage.

words, a user is likely to avoid using AC throughout the day only if his AC usage rate on other days are low. The 'low-AC usage days' is defined as the days with their AC usages account for less than T_1 of the daily period. The 'reluctant AC user' is labeled if his/her 'low-AC usage days' account for more than T_2 of the measuring period, excluding the 'no AC usage days'. Appendix A gives further analysis by the daily AC operation rates to support the assumption. Fig. 4 plots the number of reluctant AC users for different T_1 and T_2 threshold settings. A lower number of such reluctant AC users were found for lower T_1 and higher T_2. Theoretically, a reluctant AC user should correspond to low T_1 for most of the monthly period. In this connection, T_1 and T_2 were set as 0.4 and 0.6, respectively, which gives 42 reluctant AC users. This was comparable to the 102 – 20 – 42 = 40 users whose 'no-AC-usage days' were regarded as the residents being away, and were excluded from calculation. In total, we collected 2,185 and 751 daily schedules during weekday and weekend periods, respectively.

2.3. Data pre-processing

On/off data series throughout the month were summarised as 'constructed variables' that were used for depicting occupant AC operational features. These variables included average on/off actions per day, the duration of each operation, and AC 'run-over-night' rates. AC units were deemed to be in the run-over-night stage if they were turned on for 95% of the time between 23:00 and 6:00. Table 1 describes the constructed variables and their definitions. The overnight rate was the percentage total days when AC units ran throughout the night (excluding days with no on/off operations). We separated all data into weekday (i.e. Monday through Friday) and weekend (Saturday and Sunday) periods and calculated the indices accordingly. AC operation on Friday nights contributed to weekend AC run-over-night rates because occupants were typically off-work and in states of relaxation.

Daily on/off actions were used to identify whether bedroom AC controls were operated many times during the day. This was essential for identifying occupants who were in their bedrooms for many short periods. High on/off action frequency indicated discontinuous AC usage, thus leading to frequent energy demand fluctuations. Days without on/off actions were excluded because occupants were considered to be away. Fig. 5 shows that these occurrences increased from 0 to 2 and 2.5 times for both weekday and weekend, respectively, and reduced for higher frequencies. AC on/off actions were 1.5–2.5 times for most AC users, and the occurrence rates were 67% for weekdays and 53% for weekends. It is presumable that many occupants turned on their AC units before sleeping at night and turned them off after waking because of the hot and humid local climate. The peak frequency, however, was found at 1.5–2 on/off times for weekdays, and 2–2.5 times on weekend days. The occurrence frequencies were more scattered on weekends than that on weekdays, thus indicating that occupant behaviour was less regular on weekdays.

AC operation duration data provided a straightforward way of identifying AC usage intensity. These results should be more useful than strictly looking at energy consumption rates because the latter can be affected by outdoor and building-envelope features (e.g. window orientation and outdoor climate). Monthly AC operation lengths were divided by the total number of turn-on actions over the same period, thus providing the average running time for each AC operation cycle. Fig. 6 shows similar occurrence profiles for weekdays and weekends, which peaked at 8–12 h, and 6–10 per AC operation for weekdays and weekends, respectively. Hours per operation ranged from 4 to 12 on both weekdays and weekends for most AC users.

Run-over-night probability was used to determine occupant willingness to use the AC. Here, AC operation likely indicated when bedrooms were occupied. Occupants tended to turn on bedroom AC in the evening to alleviate the cooling load that took time to transfer through the envelope beginning in the late afternoon. However, some occupants may have turned off their AC before or soon after sleeping (using timers). This is because some regard AC operation while sleeping as 'unhealthy'. This overnight feature was thus an effective way to identify occupants who required less cooling. Fewer user operations during the sleeping time may result in long overnight AC running periods that use a great deal of energy. Outdoor overnight air temperature can even be lower than indoor temperatures, thus resulting in building envelop design needs and focuses that are different from those related to daytime needs. Fig. 7 shows that the occurrence frequencies of AC run-over-night probabilities for weekdays and weekends were similar, indicating that the occupant behaviour was generally consistent. Bedrooms with AC run-over-night rates lower than 0.1 indicated that occupants rarely used AC during night-time sleep, which was notably higher on weekends than on weekdays. This seemed to implicate that overnight AC usage was relevant to workloads, pressures, or the schedules for a few occupants.

3. Results

In this section, the 102 occupants were clustered into three typical users that reflect different AC usage schedules on weekdays and weekends (3.1). The typical AC usage schedules were interpreted by the features of the building and the occupants, such as the room area, total floors and occupant ages, using the approach

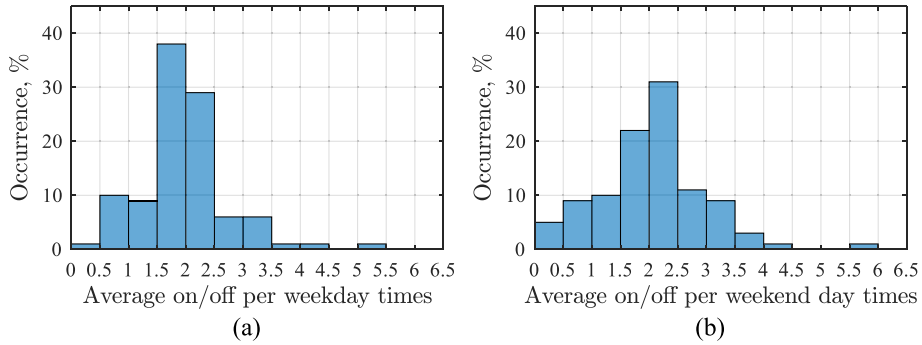


Fig. 5. Average AC on/off times per day for (a) weekdays and (b) weekends.

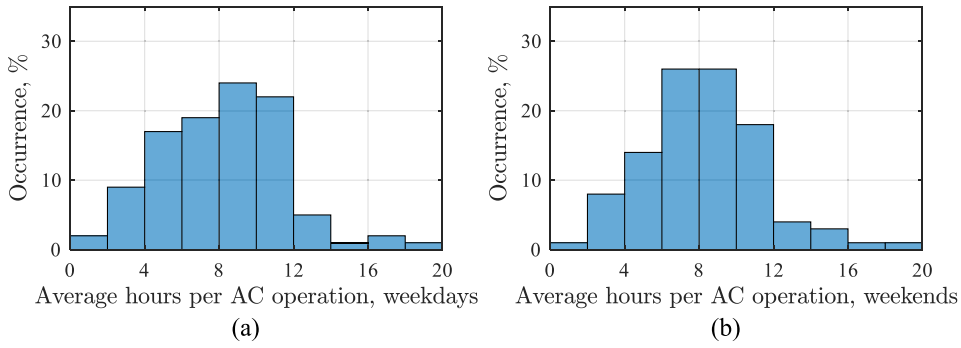


Fig. 6. Average AC operation (hours) per day for (a) weekdays and (b)weekends.

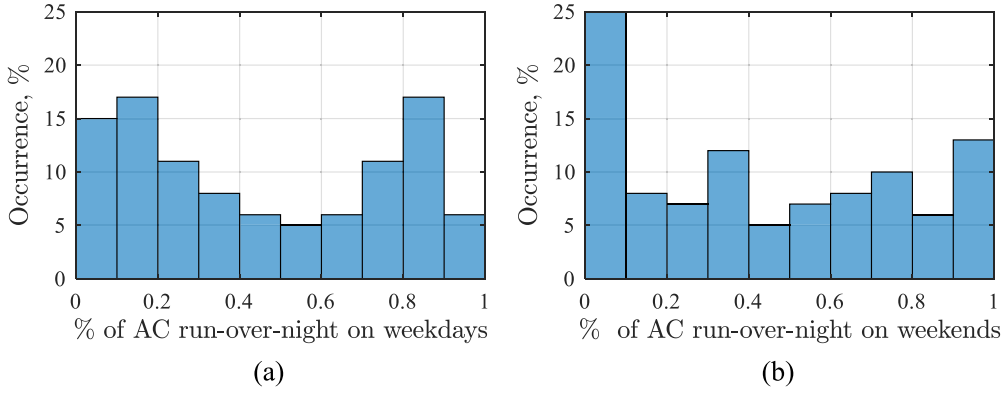


Fig. 7. AC run-over-night probabilities for (a) weekdays and (b) weekends.

of the ensembled decision Trees (3.2). The AC operation probability were converted to the AC on/off states according to the simulated thermal conditions (3.3).

3.1. Clustering representative users

The hierarchical method was used to cluster 'similar' datapoints [32]. This study determined similarity according to the Euclidean distance between data points among the three dimensions that were defined by AC operation indices. Data were rescaled to ranges of 0–1 before clustering. Each datapoint was regarded as a cluster in the first step; clusters were then merged. Here, clusters A and B to be merged should have the least distance between them; we used the average distance between object (x_{Ai}, x_{Bj}) pairs in the two given clusters according to Eq. (1) below:

$$d(A,B) = \frac{1}{n_A n_B} \sum_{i=1}^{n_A} \sum_{j=1}^{n_B} \mathrm{dist}\left(x_{Ai} - x_{Bj}\right) \tag{1}$$

Here, dist represents Euclidean distance, while n_A and n_B represent the number of data points in clusters A and B. The cluster repeats until the number of remaining clusters reaches the designated quantity or the distance increases significantly.

Fig. 8 shows a plotting of the clusters for all 102 bedrooms in addition to the percentage of each cluster according to the related AC operation feature indices. Fig. 9 shows the probability of AC usage for each cluster at different periods of the day; the data was in recorded in 1 min interval, and were averaged on an hourly basis and rounded to the nearest tenth. Our analysis revealed a total of three clusters (i.e. C1 notable noon usage, C2 significant overnight

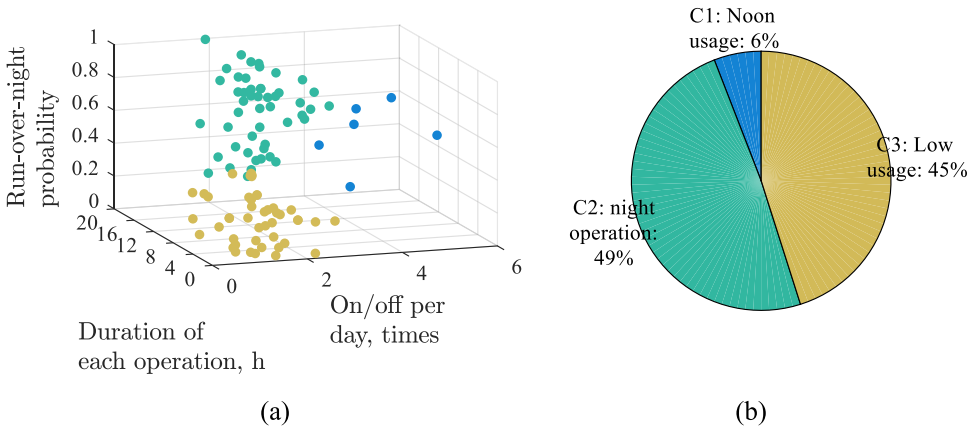


Fig. 8. (a) Cluster of all 102 cases based on weekday usage and the (b) percentages of each cluster.

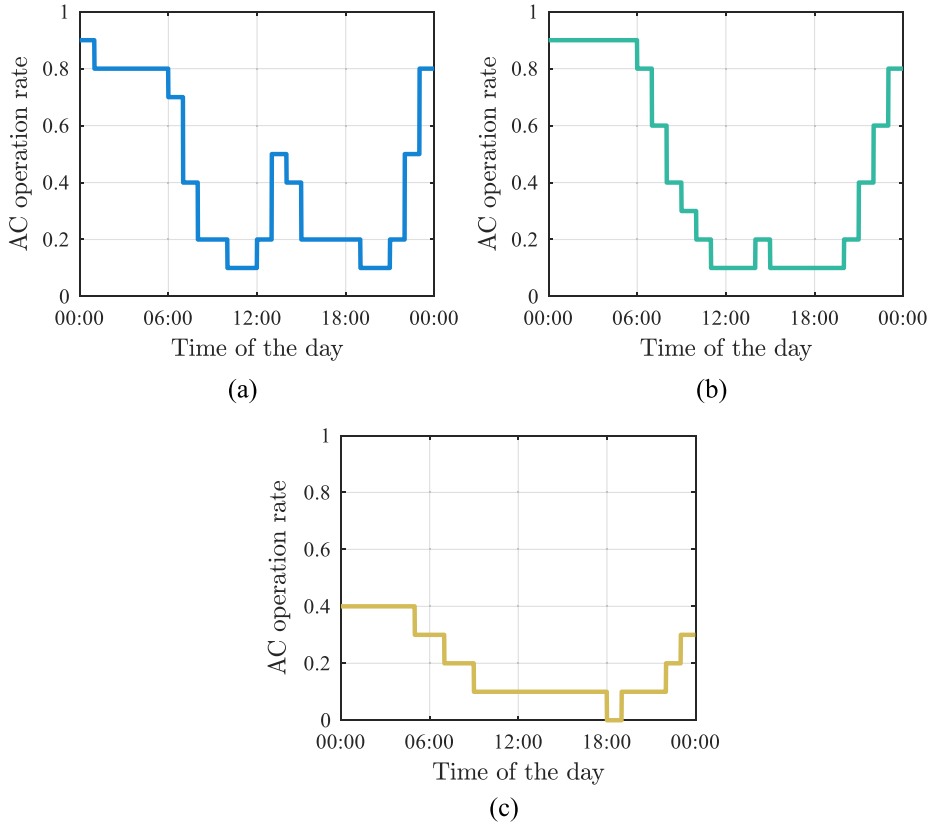


Fig. 9. Weekday AC usage rates among different representative users: (a) AC operation at noon, (b) significant AC operation at night, and (c) low AC usage.

AC usage, and C3 low daily usage). Fig. 8(b) shows that the two clusters C2 and C3 accounted for 49–45% of the total. As seen in Fig. 8(a), there were around two on/off operations per day in cases of these clusters. These clusters, according to Fig. 9(b), were either defined by a significant (up to 0.9) or relatively low (no greater than 0.4) night-time AC operation from 23:00 to 7:00, and few daytime usage from 9:00 to 21:00. The AC usage rate reductions in the morning and increase at night correspond to two on/off operations.

For the cases of C2, as given by Fig. 7(b), the likelihood of AC operation was around 90% at night, but only around 10% during daytime hours. The high likelihood of night usage of cluster C2 was in good agreement with the run-over-nigh probability greater than 0.4 for most cases as given in Fig. 8(a). This indicates behavioural features related to office workers and primary or secondary school students, who most often spend daylight hours outside the bedroom. The low usage cluster C3, according to Fig. 8(a), was generally defined by an overnight usage probability of less than 0.3 and operation durations less than 10 h, and there were around two on/off times per day (similar to the night-time operation cluster). As seen in Fig. 9(c), AC usage rates were around 0.2–0.4, and 0.1 during most of night and daytime hours, respectively, and the difference between the night-time and daytime usage rates were less than the cluster C2. These findings indicate that occupants ran their AC units for shorter periods each day and were likely to turn them off before or during overnight sleeping periods (either man-

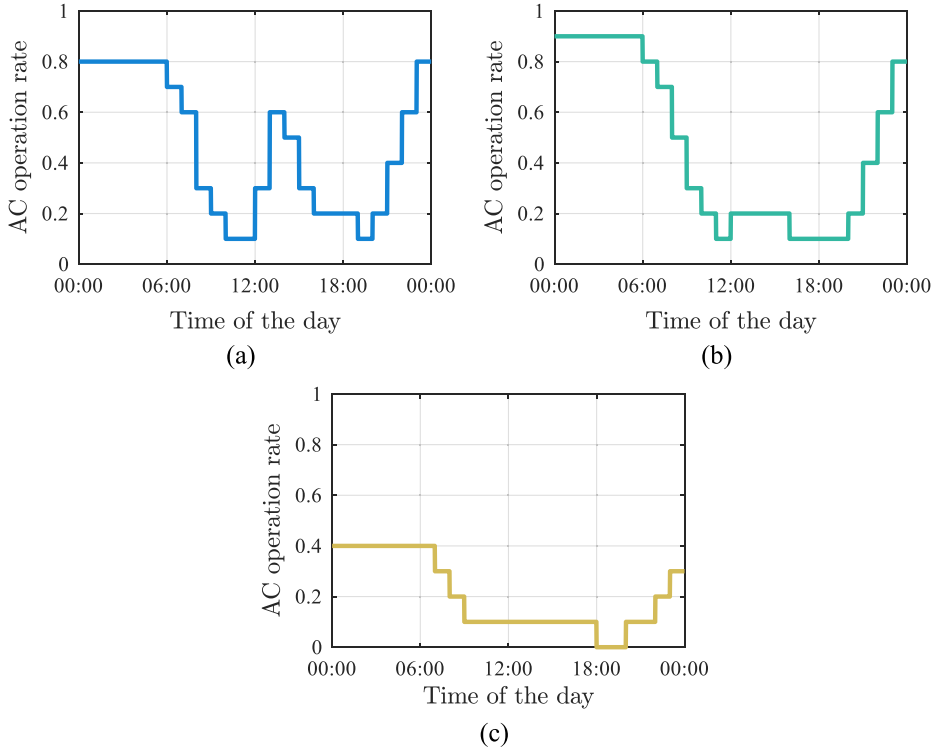


Fig. 10. Weekend AC usage rates according to different user preferences: (a) AC operation at noon, (b) significant AC operation at night, and (c) low AC usage.

ually or using timers). This profile depicts occupants who ran their AC units both discontinuously and conservatively. In addition, as seen in Fig. 8(a), there were more than 3 daily on/off times for the cluster accounting for around 6% of the total sample. Fig. 9(a) shows that this cluster was defined by around 50% AC operation rates at noon. In addition, the low usage rates from 10:00 to 12:00 and 19:00 to 21:00 were probably the result of occupants having lunch and dinner in their dining rooms. A night-time usage rate of cluster C1 was 0.8, which was comparable to that of the 'night operation' cluster C2. Users in the noon-operation cluster were likely those who took bedroom naps at that time (either during their lunch breaks from work or other spaces within the home area (e.g. living room)).

Fig. 10 shows a plotting of the AC operation probability schedules on weekends according to the clusters shown in Fig. 8. Surprisingly, the weekend profiles for each cluster were similar to those for weekdays even though they were determined by weekday AC operation indices. This similarity indicates that occupant behaviours were highly consistent on weekdays and weekends. Fig. 10(a) shows an AC probability rate of 0.6 from 13:00 to 14:00, which was higher than the operation rates during the late morning (10:00–12:00) and early evening (19:00–20:00) for all clusters. In addition, these rates were higher than those for the same periods on weekdays. For the cluster given in Fig. 10(b), the operation rate of weekends were similar to that on weekdays except a few hours in the afternoon. Fig. 10(c) shows comparable AC usage rates on weekends and weekdays as well as slight rate increment from 5:00 to 8:00 when compared to weekdays. This is seems to indicate that these reluctant AC users were more willing to use AC on weekend nights.

3.2. Representative occupant and building features

The variables for clustering cannot be accessible during the building design stage when the information about the building and its future occupants are limited. To picking up the schedule that best fits the building (or its areas)s in the design stages, the occupant schedules should be correlated with the 'explicit' occupant and building features (e.g. room size, occupant age, ...). In current work, this was accomplished by the ensemble method. The ensemble model consists of many Classification Tree [33] models developed by bootstrapped samples [34] (i.e. they are sampled uniformly and with replacements) and that estimate the cluster ID by themselves. Each Classification Tree focuses on a portion of the sample. The results of each Tree are then combined by voting. Each Classification Tree contains a series of binary partitions (splits) to classify the data into several groups. Each split should minimise the size-weighted impurity of the cluster ID in their child groups according to the threshold value of a given variable. Eq. (2) (below) gives the Gini index (G) that quantifies the impurity, where $p(k)$ is the percentage of datasets as cluster k at the node. Eq. (3) (below) gives the size-weighted reduction when it is split from parent group A into the child groups B and C. The algorithm may continuously add new Trees and splits until it reaches the designated complexity. An ensemble that uses too many Trees and splits can become excessively complicated and overfitted; trial and error tests optimise indices of this type. The importance of a variable is quantified as the sample size-weighted Gini index reduction due to the variable in all splits of the ensembled model.

$$G = 1 - \sum_{k} [p(k)]^2 \tag{2}$$

$$\Delta G = G_A N_A - G_B N_B - G_C N_C \tag{3}$$

Fig. 11 shows the 10-fold cross-validation misclassification rates of the Decision Tree ensemble according to different complexities. This ensemble correlates the representative occupants (i.e. cluster ID) with both building and occupant features. This model development was repeated 100 times to achieve an average performance. The ensemble consisted of 1–100 Decision Trees with the

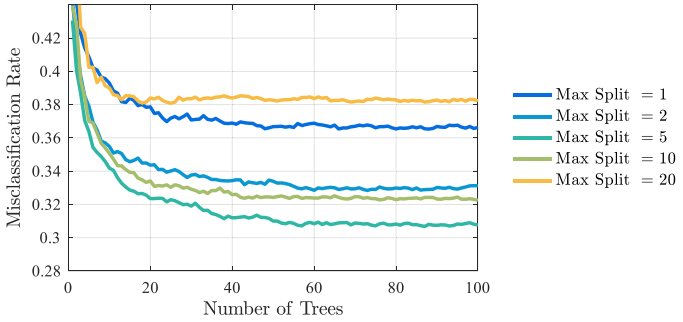


Fig. 11. Misclassification rates for the ensemble according to different complexities.

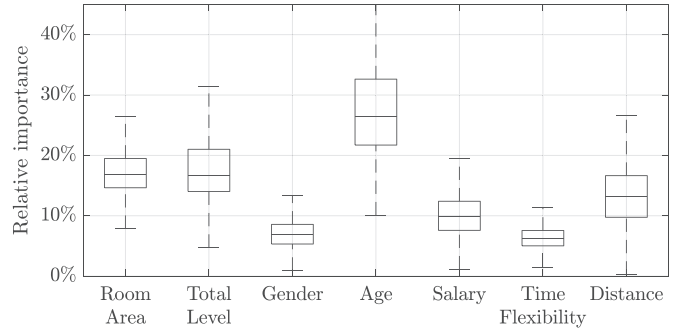


Fig. 12. Variables of relative importance for identifying representative users (i.e. cluster ID).

max splits of the Trees set as 1, 2, 5, 10, and 20. Fig. 11 shows a notable misclassification rate of up to 31%, which implicates the difficulty of correlating the occupancy schedules with the occupant and building features. In most cases, the misclassification rates were reduced when the number of Trees increased from 1 to 40; this levelled afterward, thus indicating that the model first improved when its complexity increased, but was then overfitted. Cases with minimal errors were developed by setting the max split as 5 and using more than 60–80 Trees. We used the bootstrapping approach, thus assuming that a random portion of the data were available. The relative importance of each variable, determined by the structure of the ensembled Classification Trees, was normalized as its ratio to the sum of the importance of all variables, and the results of 1000-time repeats are given in Fig. 12. Occupant age was found to be the most crucial variable for identifying representative occupant behavioural types. For instance, it indicated free time and the need for AC based on different metabolism rates. Notably, room size and total building levels were of greater importance than salary, gender, and distance from work. The total number of building levels and room sizes indicate building ages in quickly developing regions. These variables affect apartment costs and also implicate energy expenditure budgets and user preferences. The low importance of timetable flexibility likely indicated timetable uncertainty rather than indoor period preferences.

Fig. 13 shows the occurrence frequencies for key parameters related to the night-time and low-AC usage clusters on weekdays, thus indicating AC usage preferences for occupants with different features. These parameters were selected because of their high importance in identifying proper schedules. They also provided good accessibility at the design stage when conducting energy saving evaluations. The occurrence frequency of the noon-operation cluster was not plotted because of its low sample quantity. Fig. 13(a) shows that room sizes ranged from 6 to 18 m^2 for most night-time AC usage occupants. This was lower than the 12–21 m^2 for most occupants with low AC-usage rates. Such room size 'preferences' implicated occupants of a compact residential space could be more likely to stay in their bedrooms when compared to large-home users. The occurrence of the night use cluster peaked at a room size of 12 m^2 by around 35%, which can indicate a relatively low bedroom area based on local conditions. The occurrence of the low use cluster, on the other hand, peaked at room sizes of 12 and 21 m^2. Fig. 13(b) shows that total building floors peaked at five for more than 50% of the low AC-usage samples, while this occurrence was more evenly distributed (from 5 to 15) for the samples defined by night-time AC usage. Apartment height indicated building age in quickly developing regions, and probably the user expenditure preferences on the AC electricity costs. Fig. 13(c) shows the occupant age occurrence for the differently scheduled representative occupants. In the case of the night-time user cluster, occupant age occurrence peaked at either 20 or 50 years old for 26% and 20% of all samples, respectively. The low-usage cluster peaked by 26% at 30 years of age). This implicate the long-term outdoor activity of the occupant aged by 30 or their notable financial burdens. Similar features of the occurrence frequency for different clusters were found on weekends.

3.3. AC on/off schedules

The schedules shown in Figs. 7 and 8 indicate the AC usage probability. This, however, cannot be used in routine building energy simulations directly, since a deterministic 'on' or 'off' condition over the schedule is needed. Therefore, the AC operation probability should be converted to on/off states. A previous study set the probability of 50% as the threshold to turning on the AC [31]. However, we considered that an arbitrary setting could lead to high uncertainty, since the occupant sensations on the thermal environment were not considered. In this connection, we simulated the thermal comfort levels of the typical bedrooms by turning on the AC when the operation probability reaches a threshold. The threshold was set as 0.2, 0.3, …, 0.7; and the occupancy-weighted hourly average Fanger Predicted Percentage of Dissatisfied [35] for each threshold setting, as defined in Eq. (4), were simulated. The finalized threshold is thus determined to satisfy the thermal needs of most occupants.

$$\mathrm{PPD}_{W,avg} = \frac{\sum_{i=1}^{N} n_i(\mathrm{PPD}_i)}{\sum_{i=1}^{N} n_i} = \frac{\sum_{i=1}^{N} r_i(\mathrm{PPD}_i)}{\sum_{i=1}^{N} r_i} \tag{4}$$

Here, r_i represents the rate of occupancy in hour i, while PPD_i in hour i is determined by EnergyPlus simulations. The bedroom-occupant probability for a given hour was assumed to be equal to the AC operation rate during the hot and humid summer months in Guangzhou. n_i is the equivalent indoor occupant capita at hour i. N stands for total hours (the simulation was run from 1 to 31 July to represent a typical cool month). Hours with low occupancy rates were of lower weight to the $\mathrm{PPD}_{W,avg}$ than those with higher rates. The $\mathrm{PPD}_{W,avg}$ should be low if many occupants are satisfied with their indoor environment.

The bedrooms used in PPD evaluations should represent the typical design features of the residential building. We assumed a room area of around 15.5 m^2 according to an average determined for all 102 bedroom samples investigated in this study. Only one window was assumed for each room based on the 86 we determined among the 102 samples. Room length, width, and height were set at 4.2 m, 3.6 m, and 3 m, respectively; these were proper dimensions for furniture placement and space uses. A common multiplier of 300 mm was determined for these dimensions according to typical brick size and for the easy mass production of prefabricated building components. Thermal comfort levels of bedrooms with windows facing the four cardinal directions were individually evaluated. Windows were placed on the 3.6 m wall to

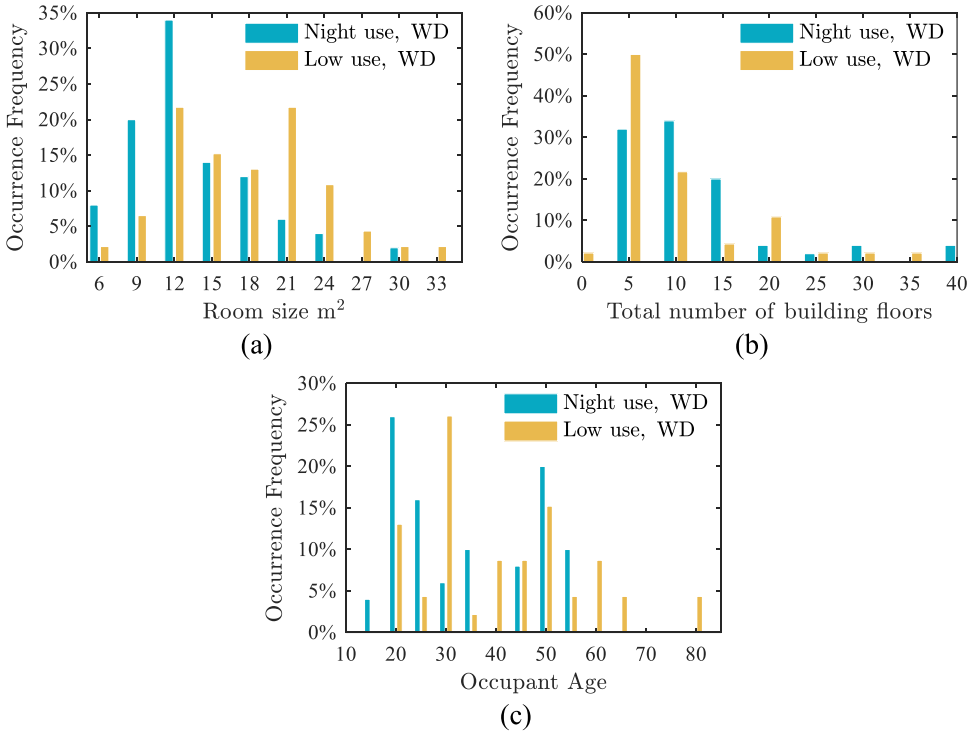


Fig. 13. Occurrence frequencies for (a) room area, (b) building height, and (c) occupant age for weekday clusters (the 'noon operation' cluster was not given because of a low sample quantity).

allow more rooms to share outside access via fenestrations. Bedrooms were assumed to have two external walls for more heat gain; it was thus more difficult to reach the thermal neutral than in units with only one external wall condition.

Fig. 14(a) and (b) show that the second external wall was assumed to face east to provide more heat gain during mornings because room occupancy rates were higher at those times compared to afternoons. The second external walls of east- and west-facing bedrooms seen in Fig. 14(c) and (d) were set on the north side. This is because the southern envelopes (as the first choice for fenestrations) were interiors walls. Dummy adjacent spaces were assumed to determine indoor air temperatures without AC. All rooms were set on middle floors with adiabatic roofing and flooring. As typical designs for the local environment, the overall heat transfer coefficient of the standard double-glazed windows and cavity walls was assumed to be 3 and 1.5 W/(m^2 K), respectively. The design temperature was assumed as 26 °C. The absorptance of the outside wall was assumed as 0.7. For clear double-glazing, the Solar Heat Gain Coefficient (SHGC) was assumed as 0.6. The window-to-wall ratio (WWR) was assumed as 0.4 for north and south, and 0.3 for east and west directions according to the upper limits of local codes [16]. This avoided thermal statutes taking advantage of low sunlight heat gains. The WWR upper limits were in good agreement with an investigation of the local residential buildings [36]. Indoor air-flow speed was assumed at 0.2 m/s to provide a typical value for domestic split-type AC usage. Occupants were assumed to use 0.5 clo (casual summer clothing). Occupant metabolic rate was assumed at 108 W/m^2 (a seated and quiet position).

Fig. 15 shows occupancy-weighted average PPD for different AC probability thresholds for different schedules. The horizontal axis represents the AC operation rate threshold; the air-conditioner was turned on if the AC operation rates, as shown in Figs. 7 and 8, were no lower than the threshold. The $PPD_{W,avg}$ of the rooms facing different directions were averaged because their differences were lower than 3.5%. The same figures show that turning on the AC at a low operation rate always more significantly reduced discomfort than at higher rates. The $PPD_{W,avg}$ increased from around 10% to around 25% when the threshold increased from 0.2 to 0.3 for schedules except the low usage one. As shown in Fig. 15(a), the $PPD_{W,avg}$ of noon-operation users rapidly increased from 52% to 97% when the threshold increased from 0.8 to 0.9. Night-operation users, however, seemed to tolerate a higher threshold better than noon-operation users. This is because AC operation rates among night-time users were higher than 0.9 for longer periods than those of noon users. For the low-usage cluster, the discomfort rate increased dramatically (to 35%) when the threshold increased from 0.1 to 0.2, as Fig. 15(c) shows. This is because the comparable maximum and minimum usage probabilities of the low-usage cluster.

Considering that a Predicted Mean Vote (PMV) of 1 (slightly warm) would result in a PPD of around 26%, Figs. 7 and 8 show that the AC should be turned on when its operation probability reaches 0.3. For the low AC usage occupants, the threshold was set as 0.2 for a $PPD_{W,avg}$ of 35%, since the occupants that needs less AC may tolerate warmer environment than the others. These threshold findings were slightly lower compared to the threshold of 0.5 of the previous study [31]. The threshold should also be set at 0.1 to result in a thermal neutral corresponding to a PPD of around 10% and a PMV within ±0.5. This finding probably explained the full-time/full-space AC operation settings requested by Chinese local design guides and the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) apartment guidelines. Finally, thresholds of 0.2 and 0.3 for 'slightly warm' indoor thermal performance gives the AC on/off schedules for different representative occupants, as shown in Table 2. It is found that there were almost no AC operations during the day period from 10:00 to 21:00 excluding the noon operation case. For the noon operation schedule, however, the AC unit were in operation from 13:00 to 15:00 on weekdays and 12:00 to 16:00 on weekends for

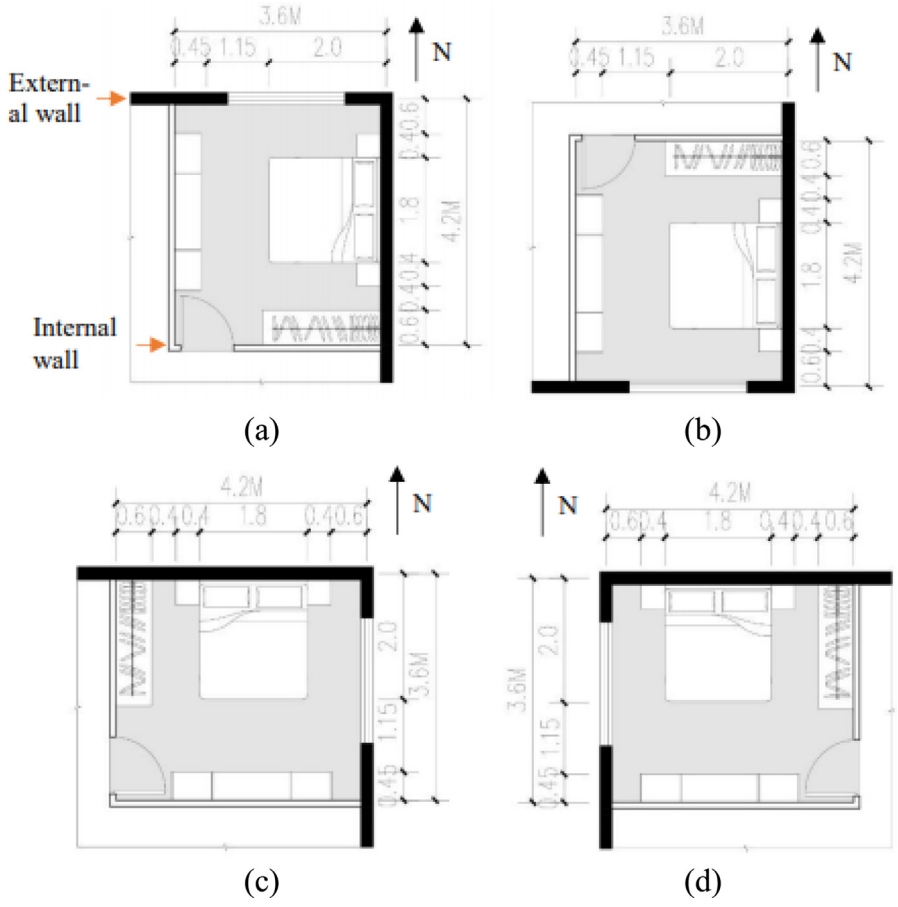


Fig. 14. (a) North, (b) south, (c) east, and (b) west-facing room layouts for thermal comfort evaluations.

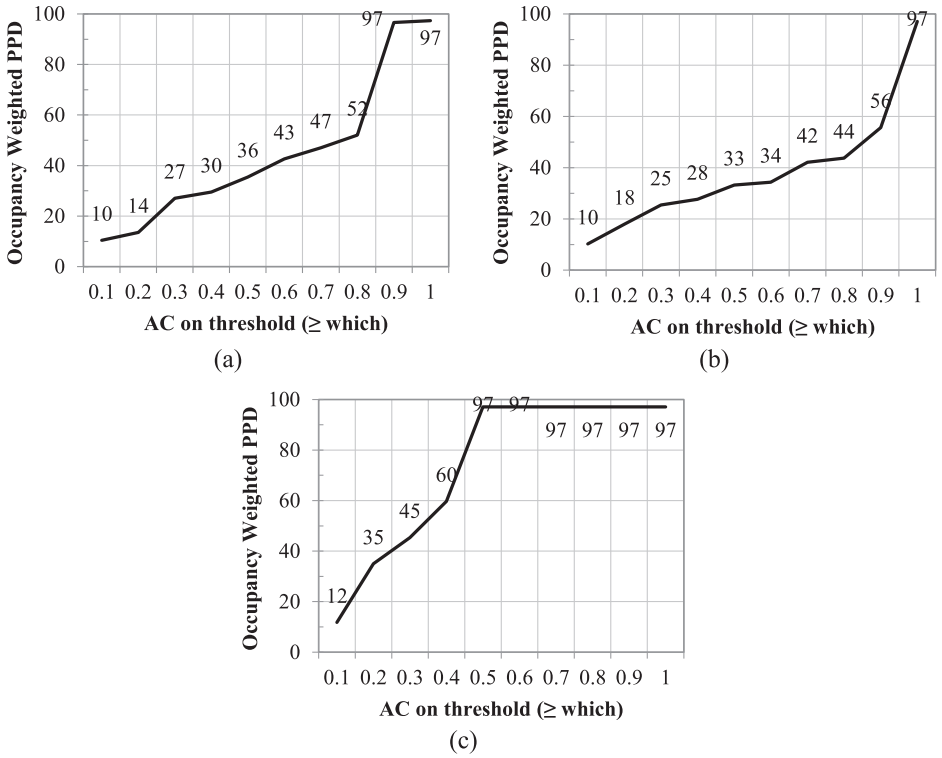


Fig. 15. Occupancy weighted PPD for different operation-rate thresholds as defined by 'on' AC statuses for (a) noon-operation, (b) night operation, and (c) low-usage schedules.

Table 2
On/off air-conditioning schedules among different representative occupants.

Time	0-1	1-2	2-3	3-4	4-5	5-6	6-7	7-8	8-9	9-10	10-11	11-12	12-13	13-14	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	22-23	23-0
Week Noon	1	1	1	1	1	1	1	1	0	0	0	0	0	1	1	0	0	0	0	0	0	0	1	1
-days Night	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1
Week Noon	1	1	1	1	1	1	1	1	1	0	0	0	1	1	1	1	0	0	0	0	0	1	1	1
-ends Night	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1
Low usage	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

the occupation at noon. For the night operation case, the AC units stops working 2 h later than the noon-operation case in the morning, and start 1 h earlier at night. A comparison of the schedules of the night and noon operation schedules shows that the AC usage can be intensive in different periods of the day for different users. It is surprising to notice that the schedules of the low usage cluster were similar to that of the night usage cluster, and difference of the on/off time was 1 hour in the morning and evening. This is probably because the thermal needs and the variables related to the thermal conditions, such as air flow speed, were assumed similar for all clusters. However, occupants of the low usage cluster may take different approaches to adjusting the thermal environments for their own needs.

4. Conclusions and future works

This analysis resulted in three representative occupant behavioural clusters related to AC operation features among residential buildings in subtropical Guangzhou, South China. Here, we (1) proposed typical AC operation probability and on/off schedules for the cases that is dominated by the cooling needs; and (2) correlated the typical AC usage patterns with the user social backgrounds. Approaches were developed to systematically (1) identifying whether the missing AC on/off recordings were because of the users' willingness or not, and (2) converting the hourly AC operation probability to the on/off status of an AC unit. The findings show that the AC usage was most common at night (i.e. 49% of all occupant samples were featured with a significant AC usage probability of 90% during those hours, but a low usage rate of 10% during the day). On the other hand, a moderate AC usage rate of 40% was found during night-time hours for 45% of all occupant samples, thus indicating notable overnight AC usage uncertainties. Finally, a third cluster (6% of the total sample) was defined by high AC usage rates at night-time and noon periods (i.e. from 23:00 to 7:00 and 13:00 to 14:00, respectively). The factors most closely related to occupant behaviours were identified according to ensemble Decision Trees (including bedroom size, total number of floors, and occupant age). Finally, AC on/off schedules for residential buildings were derived according to AC operation probabilities. AC units were regarded as 'on' when operation probability was higher than the 0.2 or 0.3 thresholds, which was determined by testing the occupancy-weighted thermal comfort of typical rooms facing the four cardinal directions. These proposed occupant behaviours and AC on/off schedules are essential for conducting accurate energy evaluations among residential buildings. Simulations based on realistic behavioural patterns may result in energy-saving design focuses that are more useful than those related to the full-time/full-space approach.

The current study, however, focused on the bedroom only, and used a low sample size. The operation modes of the AC unit were not specified in detail, and the occupant behaviour can be different for the heating cases. Unlike the AC operation that can be tested by smart meters, it can be difficult to collect the user identity information and monitoring the room occupancy directly due to privacy concerns. As a result, the occupancy was not monitored directly, making it difficult to determine whether the 'no-AC-operation' was a result of the user willingness or because the user was not there. Similarly, the indoor thermal conditions were not measured as well, though the different occupants may have different temperature needs and set points, and interact with the indoor environment and the AC unit differently. Correlations between the user identity and their corresponding energy usage preferences was thus studied in a limited size and depth. Such issues should be further explored in future studies.

Declaration of Competing Interest

There are no competing interests to declare.

Funding

This work was supported by the National Natural Science Foundation of China [Grant number: 51708137]. The project title is 'Research on the renovation of facade about the existing residential building in Lingnan area based on energy track'. (in Chinese: 基于用能轨迹的岭南地区既有住宅立面更新研究)

Appendix A. The daily AC usage rates for all users

Fig. A1 plots the rate of the daily AC operation during the period, which is the ratio of the AC operation time to the time of the day. The figure shows that the AC operation rate of some occupants (e.g. 5, 6, 50, 63, 86, and 87) are higher than 0.7 (the light/yellow color) for many days. Likewise, the rate was moderate (over 0.3–0.4, greenish blue color) for many occupants such as 2, 4, 7-9, 14, 15, 18-24, and so on, for a long time. The high and medium AC usage rates during the days implicates the high demand for cooling, and there were only a few days when the operation rate is zero (e.g. Day 14–15 of occupant 7). Considering the high and consistent AC demand of such users (high usage rate for many days), it is thus unlikely for them to change their AC usage habit so strongly (i.e. to zero) in one month. Therefore, the zero AC usages during, for example, days 14–15 of occupant 7, is unlikely a result of the user willingness. For some occupants (e.g. 10, 36–38, 40, 42–45, and so on), the AC usages was consistently low throughout the month. This shows a low AC demand and usage willingness of these authors that can lead to days without any AC usages. The thresholds of 'low AC usage' day and its occurrence frequencies are quantified by T_1 and T_2 on Fig. 4, respectively. Additionally, the days without AC usage were at different time of the month for different AC users. Thus, the no-AC-usages are unlikely to be a result of the user actions in respond to the outdoor environment only. By setting $T_1 = 0.4$ and $T_2 = 0.6$, the reluctant AC users include occupants 10, 13, 25, 31, 34, 36-38, 40, 42-44, 47, 56, 59, 62, 70, 75, 78, 81, 85, 88–90, 93, 94, 99, 100, whose no-AC-usage days were deemed as the results of the user willingness.

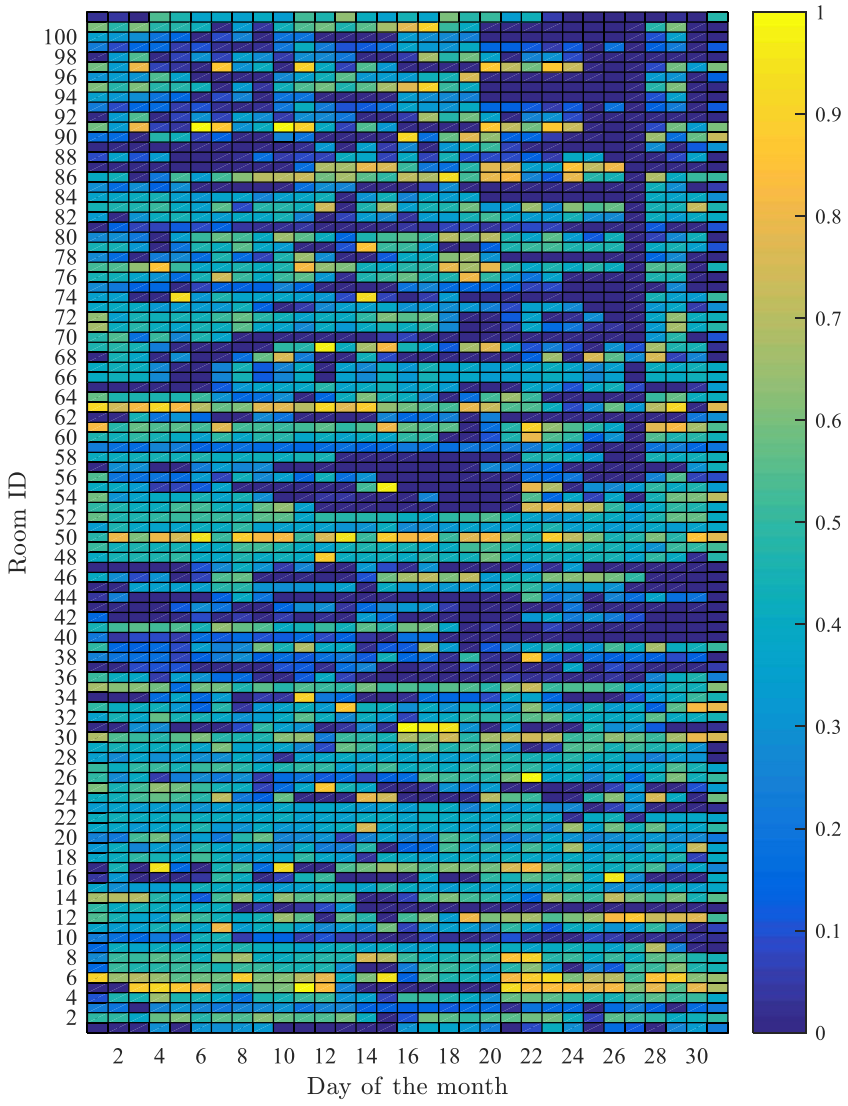


Fig. A1. The daily AC operation rate throughout the measurement period.

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